The finite-size shape transition of coin-turning walks

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Abstract

A coin-turning walk on the integers switches direction at step k with probability c/k. With a fair first step the fraction of right steps converges in law to $\mathrm{Beta}(c, c)$, which is U-shaped for $c < 1$, uniform at $c = 1$ and peaked at the center for $c > 1$. We locate this shape transition at finite N. With a fixed first step it happens exactly at $c = 1$ for every N. The number of right steps is then uniform on $\{1, \dots, N\}$, and the ratio of the mass of each interior point to that of the end increases strictly in c, so every interior point ties with the end at $c = 1$. The same holds for a walk on d states, whose count vector is uniform at the tie. Both laws come from a chain that draws its state afresh or repeats it, and we give the joint law of its counts and last state for every weight. With a fair first step the center ties with the end at $c_N = 1 - \log 2/(\log N + \gamma + 2 - 2\log 2) + O((\log N)^{-3})$, where γ is Euler's constant. The proof rests on a local limit theorem with a rate for the $\mathrm{Beta}(c, c)$ density at the center.

Keywords: coin-turning walk; time-inhomogeneous Markov chain; occupation law; Pólya urn; Beta distribution; local limit theorem.
MSC2020 subject classifications: Primary 60J10; 60F05, Secondary 60C05; 60G50.

1 Introduction

A coin-turning walk on $\mathbb{Z}$ switches direction at step $k \geq 2$ with probability c/k, independently of the past, where $0 < c < 2$. This is the walk of Engländer and Volkov [5, 9] with $p_k = c/k$. Let A_N be the number of right steps among the first N. With a fair first step A_N/N converges in law to $\mathrm{Beta}(c, c)$ [4, 5]. The limit has a shape transition at $c = 1$. Its density is U-shaped for $c < 1$, uniform at $c = 1$ and peaked at $\frac{1}{2}$ for $c > 1$. We ask where this transition happens at finite N, that is, at which c an interior point of $\{0, \dots, N\}$ becomes as likely as an end. With a fixed first step the answer is exactly $c = 1$, for every N and every interior point (Proposition 2.3). With a fair first step the center ties with the end at some $c_N < 1$, and $1 - c_N \sim \log 2/\log N$ (Proposition 2.7). The walk with d states starts in state 1, and at step $k \geq 2$ it moves to each of the other $d - 1$ states with probability $\theta/(k + d - 2)$. For $d = 2$ this is c/k with $c = \theta$. Its count vector shows the same exact transition at $\theta = 1$ (Theorem 2.4).

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1.1 Setting and notation

The walk takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, and X_N is the last step. Let $A_N = \#\{i \le N : X_i = +1\}$ be the number of right steps, and write $P_N(k) = \mathbb{P}(A_N = k)$ for the mass of the point k. We write $\mathbb{P}_+$ for probability given $X_1 = +1$. With a fair first step, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and $P_N(k) = P_N(N-k)$. In that case, for even N, we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. The points $1, \dots, N-1$ are interior. A crossing is a value of c at which an interior point ties with an end. Logarithms are natural, $H_k = \sum_{j \le k} 1/j$, and γ is Euler's constant. Limits and $O(\cdot)$ refer to $N \to \infty$, over even N when C_N appears.

1.2 Results and method

With a fixed first step and $c = 1$, A_N is exactly uniform on $\{1, \dots, N\}$. The ratio of the mass of each interior point to that of the end increases strictly in c. Hence every interior point ties with the end exactly at $c = 1$, for every $N \ge 2$ (Proposition 2.3). With d states and $\theta = 1$ the count vector is exactly uniform over the compositions $(n_1, \dots, n_d)$ of N with $n_1 \ge 1$, and every other count vector ties with the corner $(N, 0, \dots, 0)$ exactly at $\theta = 1$ (Theorem 2.4). Both laws come from a chain that draws its state afresh or repeats it. Its counts and last state have an explicit joint law for every weight (Proposition 2.1). With a fair first step each interior point is twice as likely as an end at $c = 1$, and its crossing is unique and below 1 (Proposition 2.6). For the center $c_N = 1 - \log 2/(\log N + \gamma + 2 - 2\log 2) + O((\log N)^{-3})$ (Proposition 2.7, proved in Appendix A).

The exact laws come from an induction that carries the last state along with the counts. The ties come from writing each ratio of an interior point to the end as a sum over paths of products of odds that increase in c. For the fair start we couple the walk with the chain that draws afresh or repeats, whose counts are a mixture of binomial laws. Summation by parts turns the coupling into a local limit theorem with a rate at the center (Proposition A.3).

Verification. No proof rests on a computation. Two scripts, verify_aging.py and verify_aging_urn.py, recompute the numbers of this note, and figures/make_aging_figure.py draws Figure 1. In exact arithmetic $C_{200} - E_{200}$ changes sign between $c_{200} - 10^{-6}$ and $c_{200} + 10^{-6}$. Every proposition, the lemma and the theorem are also proved in Lean 4 (version 4.33.1) with Mathlib at commit `0df444a360ea`, in kagey-problems/problem131/lean/PaperD of https://github.com/apemm/Kagey-Problems, using only the axioms `propext`, `Classical.choice` and `Quot.sound`. The file MAP.md there lists them under the labels of this note, and in an older table under the numbers of an earlier draft. Its numbers 3.1 to 3.7 are 2.1 to 2.7 here, and its numbers B.1 to B.3 are A.1 to A.3 here.

1.3 Prior work

Dietz and Sethuraman [4] proved the limit law of A_N/N, Engländer and Volkov [5] gave a second proof, and Engländer, Volkov and Wang [9] found the scaling limit. For d states the limit is Dirichlet [4, Theorem 1.4], and Bouguet and Cloez [2, Propositions 4.1 and 4.5] find the joint limit with the current state. The uniform law for 2 states is an exercise in [9, Section 3.1, footnote 1]. The count law of the chain of Proposition 2.1 also follows from Dietz, Lippitt and Sethuraman [3, Theorem 2.18]. Chapter 7 of the March 18, 2024 author draft [7] of Engländer and Volkov's monograph [8] gives the two-state joint law with the last state for integer urn weights. We cite the draft for every locator in that work. Section 2 gives the precise overlaps. Pradhan, Roy and Ray [15] find, with the methods of physics, the second moment and the run tail of the walk with switch probability c/k^α, which is ours at $\alpha = 1$. What we add is the joint law for $d \ge 3$

or a weight that is not an integer, the strict ordering that turns the uniform laws into ties (Proposition 2.3(b) and Theorem 2.4(b)), and the expansion of c_N with the local limit theorem behind it. For runs of geometric length the center overtakes the ends when the expected number of switches is about $\log N$, for 2 directions [13] and for d directions [14].

2 Exact laws and ties

Recall the model. After its first step the walk switches direction at step $k \geq 2$ with probability c/k, independently of the past, where $0 < c < 2$. So the walk switches less and less often, and the expected number of switches in N steps is $c(H_N - 1)$, which is about $c \log N$. With a fair first step, A_N/N converges in law to $\mathrm{Beta}(c, c)$ [4, Example 1.1 and Theorem 1.4], [5, Theorem 2]. At finite N the first step matters. We first give one exact law behind all the ties, then fix the first step, pass to d states, and return to the fair first step.

As in Section 1.1, $\mathbb{P}_+$ is the law given $X_1 = +1$ and X_N is the last step. A path is a sequence $x \in \{\pm 1\}^N$, and it switches at time $t \geq 2$ if $x_t \neq x_{t-1}$. Step $k \geq 2$ contributes the factor c/k to the probability of a path if the path switches there, and $1 - c/k$ otherwise. So relative to no switch, a switch at time t multiplies the probability of a path by the odds

$$\omega_t = \frac{c/t}{1 - c/t} = \frac{c}{t - c} \qquad (t \geq 2),$$

and

$$\mathbb{P}_+(X = x) = \prod_{k=2}^{N} \Big(1 - \frac{c}{k}\Big) \prod_{t:\, x_t \neq x_{t-1}} \omega_t \qquad (x_1 = +1). \tag{2.1}$$

Each ω_t is positive and continuous on $(0, 2)$, increases strictly in c, and tends to 0 as $c \downarrow 0$.

2.1 Drawing afresh or repeating

All the exact laws of this section come from one chain, which works for every weight. Fix $d \geq 2$ and $\theta > 0$, and let $Y_1, Y_2, \ldots$ be the following chain on $\{1, \ldots, d\}$. At step $t \geq 1$, with probability

$$f_t = \frac{d\theta}{d\theta + t - 1}$$

the state Y_t is drawn afresh, uniformly from the d states, and otherwise $Y_t = Y_{t-1}$ (so Y_1 is uniform, since $f_1 = 1$). In other words, at step $t \geq 2$ the chain moves to each other state with probability $\theta/(d\theta + t - 1)$. This is the mechanism of the conservative random walk of Engländer and Volkov [6], who redraw a direction in $\mathbb{Z}^d$ with probability p_n and study recurrence and scaling limits. Let M_T be the vector of visit counts of $Y_1, \ldots, Y_T$, and write $x^{(m)} = x(x+1)\cdots(x+m-1)$.

Proposition 2.1 (fresh or repeat)**.** *For every $T \geq 1$, every composition m of T into d nonnegative parts and every state j,*

$$\mathbb{P}(M_T = m,\, Y_T = j) = \frac{m_j}{T}\, Q_T(m), \qquad Q_T(m) = \frac{T!}{(d\theta)^{(T)}} \prod_{i=1}^{d} \frac{\theta^{(m_i)}}{m_i!}. \tag{2.2}$$

So M_T has the law Q_T of the counts in Pólya's urn with d colors and initial weight θ on each color. For $d = 2$ and even $T \geq 2$,

$$\frac{\mathbb{P}(M_T = (T/2, T/2))}{\mathbb{P}(M_T = (T, 0))} = \binom{T}{T/2} \prod_{i=0}^{T/2-1} \frac{\theta + i}{\theta + T/2 + i}, \tag{2.3}$$

which increases strictly in θ and equals 1 exactly at $\theta = 1$.

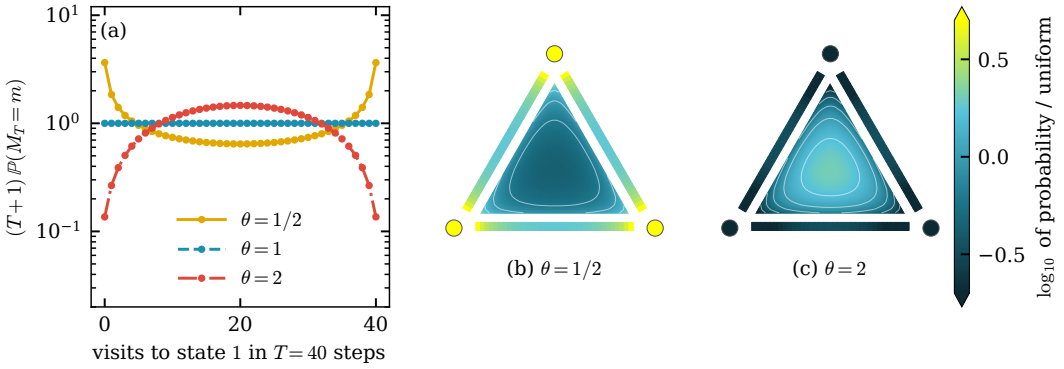


Figure 1: The exact law (2.2) of the counts of the chain that draws afresh or repeats. (a) Two states and $T = 40$, as a multiple of the uniform law. The law is flat at $\theta = 1$, the ends are the most likely points for $\theta = \frac{1}{2}$, and the center is for $\theta = 2$. (b), (c) Three states and $T = 36$ on the simplex of visit fractions, colored by the probability of a count vector relative to the uniform law. The triangle holds the count vectors that use all 3 states, the bands hold those that use 2, and the disks are those that use 1. The thin lines are level lines. For $\theta = \frac{1}{2}$ a corner is 9.6 times as likely as under the uniform law and the center 0.43 times. For $\theta = 2$ the numbers are 0.035 and 2.1.

In words, the chain repeats its last state where the urn repeats a uniformly chosen earlier draw, and at each fixed time this makes no difference to the counts or to the last state. Figure 1 shows the law below, at and above the tie.

Proof. We use induction on T. For $T = 1$ both sides of (2.2) are $\frac{1}{d}$ when $m = e_j$ and 0 otherwise. Let $T \geq 2$ and suppose (2.2) holds for $T - 1$. If $m_j = 0$ both sides vanish. Otherwise put $m' = m - e_j$. Summing (2.2) over the last state gives $\mathbb{P}(M_{T-1} = m') = Q_{T-1}(m')$. A fresh draw gives j with probability $\frac{1}{d}$ whatever the last state was, and a repeat gives j only when the last state was j (this is why the induction carries the last state along with the counts). So

$$\mathbb{P}(M_T = m,\, Y_T = j) = \frac{f_T}{d}\, Q_{T-1}(m') + (1 - f_T)\, \frac{m_j - 1}{T - 1}\, Q_{T-1}(m') = \frac{\theta + m_j - 1}{d\theta + T - 1}\, Q_{T-1}(m').$$

On the other hand $Q_T(m) = Q_{T-1}(m') \cdot \frac{T}{m_j} \cdot \frac{\theta + m_j - 1}{d\theta + T - 1}$ by the definition of Q_T. This gives (2.2). Summing over j gives $\mathbb{P}(M_T = m) = Q_T(m)$, and the ratio (2.3) is $Q_T(T/2, T/2)/Q_T(T, 0)$ with $\theta^{(T)} = \theta^{(T/2)}(\theta + T/2)^{(T/2)}$. Each factor $(\theta + i)/(\theta + T/2 + i)$ increases strictly in θ, and at $\theta = 1$ the product is $(T/2)!^2/T!$. □

The law of the counts alone is known. Dietz, Lippitt and Sethuraman [3, Theorem 2.18] show that the moments of the limit law of the chains of [4] are occupation probabilities at finite N of a second chain. In the case of a Dirichlet limit that second chain is Y, with the uniform law in the role of their μ and $d\theta$ in the role of their θ. The limit is then the Dirichlet law with every parameter θ [4, Theorem 1.4], and its moments give Q_T. Engländer and Volkov work it out in Chapter 7 of their March 18, 2024 draft [7]. For 2 states and an integer weight θ they prove (2.2), with the last state, for every T [7, Theorem 7.2, pp. 138–139], and they credit this result to E. Crane. What the proposition adds is the joint law with the last state for $d \geq 3$ states, and for a weight θ that is not an integer. As $T \to \infty$, (2.2) turns into the joint limit of the fractions of time and the current state found by Bouguet and Cloez [2, Proposition 4.1], in which the Dirichlet law is size-biased by the coordinate of the current state.

Remark 2.2. Here is another way to see the law of the counts in Proposition 2.1 (but not the joint law with the last state). The fresh draws cut $Y_1, \dots, Y_T$ into runs, and each run gets a uniform state. The urn can be run in the same way, as a Chinese restaurant process in which each table gets a uniform color. There a table is opened with the same probabilities f_t. The lengths of the runs are the spacings between the successes of independent trials with success probabilities f_t. It is a classical fact (the Feller coupling, see [1]) that the multiset of these spacings has the same law as the multiset of the table sizes (the two ordered lists do not). The proof in [7, Section 7.4], like Proposition 2.1, instead goes through the last state. We checked (2.2) in exact arithmetic for 4 values of θ, with $T \le 7$ for $d = 2, 3$ and $T \le 5$ for $d = 4$ (`verify_aging_urn.py`).

2.2 A fixed first step

Here the walk starts with a step $+1$.

Proposition 2.3 (2 states, fixed first step)**.** *Let $N \ge 2$ and let the first step be $+1$.*

(a) At $c = 1$, for $1 \le s \le N$,

$$\mathbb{P}_+(A_N = s,\, X_N = +1) = \frac{s-1}{N(N-1)}, \qquad \mathbb{P}_+(A_N = s,\, X_N = -1) = \frac{N-s}{N(N-1)}.$$

So A_N is uniform on $\{1, \dots, N\}$.

(b) For $1 \le s \le N-1$ the ratio $\mathbb{P}_+(A_N = s)/\mathbb{P}_+(A_N = N)$ increases strictly in $c \in (0, 2)$, and it equals 1 exactly at $c = 1$.

For $c < 1$ the end N is the most likely value of A_N, and for $c > 1$ it is the least likely of the values $1, \dots, N$ (the value 0 has probability 0).

Proof. (a) At $c = 1$, the shifted chain $Y_t = X_{t+1}$ is the chain of Proposition 2.1 with $d = 2$, $\theta = 1$ and uniform first state. Apply (2.2) at $T = N - 1$ with counts $(s - 1, N - s)$, where $Q_{N-1} = 1/N$, then sum over the last state.

(b) We write the ratio as a sum over paths. Only the path with every step $+1$ has $A_N = N$, and it never switches. So $\mathbb{P}_+(A_N = N)$ is the first product in (2.1), and dividing by it leaves

$$\frac{\mathbb{P}_+(A_N = s)}{\mathbb{P}_+(A_N = N)} = \sum_x \prod_{t:\, x_t \ne x_{t-1}} \omega_t, \tag{2.4}$$

where the sum is over the paths x with $x_1 = +1$ and s steps equal to $+1$. For example, for $N = 3$ and $s = 2$ the paths are $(+, +, -)$ and $(+, -, +)$, and the ratio is $\omega_3 + \omega_2\omega_3$. For $1 \le s \le N - 1$ there is such a path, and each of them switches at least once. So the right side is a sum of products of the ω_t with at least one factor each. Each ω_t is positive and increases strictly in c, so the sum does too. By (a) it equals 1 at $c = 1$. □

After a shift of one step, part (a) is the exercise in [9, Section 3.1, footnote 1]. That footnote also says that the fact can be related to Pólya urns (the authors credit M. Balázs) and that a more general connection of their model with Pólya urns will appear in a later paper. That connection is developed in Chapter 7 of the dated draft, and after the same shift part (a), with the last state, is [7, Claim 7.1]. So part (a) is not new. What we add for 2 states is part (b), and the count law also follows from [3, Theorem 2.18] (Section 2.1). Both parts of Proposition 2.3 are also proved in Lean (Section 1.2).

2.3 d states

The same thing happens for a chain on d states that leaves its current state for each other state at the same aging rate. Fix $d \geq 2$ and θ with $0 < \theta < d/(d-1)$. The chain $X_1, X_2, \dots$ on $\{1, \dots, d\}$ starts at $X_1 = 1$. At step $k \geq 2$ it moves to each of the other $d-1$ states with probability $\theta/(k+d-2)$, and it stays with probability $1-(d-1)\theta/(k+d-2)$. The condition on θ says that the probability of staying is positive at every step $k \geq 2$. Let $L_N = (L_N(1), \dots, L_N(d))$ be the vector of visit counts, namely $L_N(i) = \#\{k \leq N : X_k = i\}$. Its values are the compositions $n = (n_1, \dots, n_d)$ of N into d nonnegative parts with $n_1 \geq 1$. There are $\binom{N+d-2}{d-1}$ of them (subtract one from n_1). The corner Ne_1, where e_i is the i-th unit vector, is the count vector of the chain that never moves.

For such chains Dietz and Sethuraman [4, Theorem 1.4] proved that the fractions of time converge in law to the Dirichlet law with every parameter θ, which is uniform on the simplex exactly at $\theta = 1$. Our step k uses their transition matrix at time $n = k+d-2$, and the two agree for $n \geq d+1$. Both chains redraw a uniform state with probability $d\theta/n$. After the first common redraw they agree, so their fractions of time have the same limit. At $\theta = 1$, part (a) of the theorem below is an exact finite-N form of this limit.

Theorem 2.4 (d states)**.** *Let $d \geq 2$, $0 < \theta < d/(d-1)$ and $N \geq 2$.*

(a) At $\theta = 1$, for every count vector n and every state j,

$$\mathbb{P}(L_N = n,\, X_N = j) = \frac{n_j - \mathbf{1}\{j=1\}}{(N-1)\binom{N+d-2}{d-1}}. \tag{2.5}$$

So L_N is uniform on its $\binom{N+d-2}{d-1}$ values, the corner included.

(b) For every count vector $n \neq Ne_1$ the ratio $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ increases strictly in θ, and it equals 1 exactly at $\theta = 1$.

In words, at $\theta = 1$, given the counts, the last state is distributed as the state at a uniformly chosen one of the steps $2, \dots, N$ (the forced first visit to state 1 is left out). Below $\theta = 1$ the corner is the most likely count vector, and above it the corner is the least likely.

Proof. (a) At $\theta = 1$, X_2 is uniform and $Y_t = X_{t+1}$ is the chain of Proposition 2.1. Its count vector is $L_N - e_1$, and $Q_{N-1} = 1/\binom{N+d-2}{d-1}$. Formula (2.2) gives (2.5), and summing over j gives the uniform law.

(b) A path $x \in \{1, \dots, d\}^N$ with $x_1 = 1$ has probability $\prod_{k=2}^{N}\bigl(1 - \frac{(d-1)\theta}{k+d-2}\bigr)$ times the product of the odds

$$\omega_t^{(d)} = \frac{\theta/(t+d-2)}{1-(d-1)\theta/(t+d-2)} = \frac{\theta}{t+d-2-(d-1)\theta}$$

over the times $t \geq 2$ with $x_t \neq x_{t-1}$. The denominator is positive and decreases in θ, so each $\omega_t^{(d)}$ increases strictly in θ. Only the constant path has counts Ne_1. So $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ is the sum of the products of the odds over the paths with counts n. For a count vector $n \neq Ne_1$ there is such a path, and each of them moves at least once. Hence the ratio increases strictly in θ, and by (a) it equals 1 at $\theta = 1$. □

Remark 2.5. For $d = 2$ the chain switches at step k with probability θ/k, so Theorem 2.4 with $\theta = c$ is Proposition 2.3, and (2.5) reads $(s-1)/(N(N-1))$ for $j = 1$ and $n = (s, N-s)$.

By Proposition 2.1, $(L_N - e_1, X_N)$ has the law of the counts and last color of $N-1$ draws from Pólya's urn with one ball of each color. This is an identity at each fixed time, not an identity of path laws. For example, with $d = 2$ and $N = 4$ the chain path

$1,1,2,2$ has probability $1/8$, while the urn draws $1,2,2$ have probability $1/12$. For 2 states Engländer and Volkov call this relation mimicking [7, Definition 7.1 and Lemma 7.1], and they show with an example of 3 steps that the coin-turning walk and the urn have different path laws [7, Example 7.1]. Also, at $\theta = 1$ the chain moves at step k with probability $(d-1)/(k+d-2)$, independently over k. This is the probability that customer k opens a new table in the Chinese restaurant process with parameter $d-1$. So the expected number of moves in N steps at the tie is $\sum_{k=2}^{N} \frac{d-1}{k+d-2} = (d-1)\log N + O(1)$, which has the same order, $\log N$, as the number of turns at the crossing from the corners to the center for d directions and runs of geometric length (Section 1.3).

We checked (2.5), and the strict ordering in (b) along the 5 values $\theta = \frac{1}{2}, \frac{9}{10}, 1, \frac{11}{10}, \frac{5}{4}$, in exact rational arithmetic for $d = 2,3,4$ and $2 \le N \le 8$ (`verify_aging.py`).

2.4 A fair first step

We return to 2 states and let the first step be fair. Then both ends can be reached, and $E_N = \mathbb{P}(A_N = N) = \frac{1}{2}\prod_{k=2}^{N}(1 - c/k)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. So

$$\mathbb{P}(A_N = s) = \tfrac{1}{2}\big(\mathbb{P}_+(A_N = s) + \mathbb{P}_+(A_N = N - s)\big), \qquad E_N = \tfrac{1}{2}\mathbb{P}_+(A_N = N), \tag{2.6}$$

since $\mathbb{P}_+(A_N = 0) = 0$.

Proposition 2.6 (fair first step)**.** *Let $N \ge 2$ and $1 \le s \le N-1$, with a fair first step.*

(a) At $c = 1$, $\mathbb{P}(A_N = s) = \frac{1}{N}$ and $E_N = \frac{1}{2N}$.

(b) The ratio $\mathbb{P}(A_N = s)/E_N$ is continuous and strictly increasing in $c \in (0,2)$. It tends to 0 as $c \downarrow 0$ and equals 2 at $c = 1$. So there is exactly one c in $(0,2)$ with $\mathbb{P}(A_N = s) = E_N$, and it is less than 1.

For even N we write c_N for the crossing in (b) with $s = h = N/2$, so that $C_N = E_N$ exactly at $c = c_N$. In words, at $c = 1$ every interior point still has probability $\frac{1}{N}$, and the two ends share the probability $\frac{1}{N}$ that the fixed start gave to one end. Engländer and Volkov give related fair-start formulas for switch probabilities $1/(n+a)$ [7, Section 5.7, p. 76]. We derive (a) from the symmetry identity (2.6) instead.

Proof. (a) This follows from (2.6) and Proposition 2.3(a). (b) By (2.6) the ratio is the sum of the right sides of (2.4) for s and for $N - s$. Both are finite sums of products of at least one ω_t. So the ratio is continuous and strictly increasing, and it tends to 0 as $c \downarrow 0$. By (a) it equals 2 at $c = 1$. Hence it equals 1 exactly once, at some $c < 1$. □

How far below 1 is c_N? The end is of order N^{-c} and the center is of order $1/N$, so they meet when N^{1-c} is about 2.

Proposition 2.7 (the fair-start crossing)**.** *For even $N \to \infty$,*

$$c_N = 1 - \frac{\log 2}{\log N + \gamma + 2 - 2\log 2} + O\big((\log N)^{-3}\big),$$

where γ is Euler's constant.

Idea of the proof. A monotonicity argument first traps c_N within $O(1/\log N)$ of 1, so an expansion at $c = 1$ applies. The end is explicit, namely $E_N = \Gamma(N+1-c)/(2\Gamma(2-c)\Gamma(N+1))$, which is $N^{-c}/(2\Gamma(2-c))$ with relative error $O(1/N)$. For the center we prove a local limit theorem with a rate, namely $NC_N = d_c + O(\sqrt{\log N/N})$ uniformly for $c \in [c^{(3)}, 1]$, where $c^{(3)} = (9 - \sqrt{57})/2 \approx 0.725$ and $d_c = 4^{1-c}/\mathrm{B}(c,c)$ is the density of $\mathrm{Beta}(c,c)$ at $\frac{1}{2}$.

So $\log(C_N/E_N) = (c-1)\log N + \Theta(c) + O(\sqrt{\log N/N})$ with $\Theta(c) = \log(2\Gamma(2-c)d_c)$. Here $\Theta(1) = \log 2$ and $\Theta'(1) = \gamma + 2 - 2\log 2$, and solving for the zero gives the expansion. The proof is in Appendix A. There C_N is an average of explicit numbers over the law of the time of the h-th step $+1$ (Lemma A.1). The local limit theorem comes from a coupling of the walk with the chain of Proposition 2.1 with $d = 2$ and $\theta = c$, whose counts are a mixture of binomial laws with a $\mathrm{Beta}(c, c)$ parameter (Proposition A.3).

The error term decays only like $(\log N)^{-3}$, but the expansion is already accurate at moderate N. The computed crossing at $N = 200$ is $c_{200} = 0.8925$, against 0.8932 from the expansion without its error term. The constant in the error term is not computed.

A Proof of Proposition 2.7

We bound a gamma ratio, bracket C_N and c_N, prove a local limit theorem at the center, and then expand the crossing. We use the setting of Sections 1.1 and 2 with 2 states and a fair first step. So $0 < c < 2$, X_1 is a fair sign, and the switch indicators $\chi_k = \mathbf{1}\{X_k \neq X_{k-1}\}$, $k \geq 2$, are independent Bernoulli(c/k) variables, independent of X_1. Put

$$\rho(a,b) = \prod_{k=a+1}^{b} \Big(1 - \frac{c}{k}\Big) = \frac{\Gamma(b+1-c)\,\Gamma(a+1)}{\Gamma(b+1)\,\Gamma(a+1-c)}, \qquad \lambda_{t,N} = \frac{c}{t}\,\rho(t,N)$$

for integers $1 \leq a \leq b$ and $2 \leq t \leq N$. So $\lambda_{t,N}$ is the probability of a switch at time t and none in $(t, N]$, and $E_N = \frac{1}{2}\rho(1,N)$. Let T_s^+ be the time of the s-th step equal to $+1$. Let B be a Beta(c,c) variable, $d_c = 4^{1-c}/\mathrm{B}(c,c)$ its density at $\frac{1}{2}$, and $\Theta(c) = \log\big(2\Gamma(2-c)\,d_c\big)$. For even N we write $h = N/2$ and $C_N = \mathbb{P}(A_N = h)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. We call this flip symmetry.

A.1 A gamma ratio

Gautschi's inequality [12, 5.6.4] says that $x^{1-s} < \Gamma(x+1)/\Gamma(x+s) < (x+1)^{1-s}$ for $x > 0$ and $0 < s < 1$. We claim that for $0 < c < 2$ and real $y \geq 2$,

$$\frac{\Gamma(y+1-c)}{\Gamma(y+1)} = y^{-c}(1+\vartheta) \qquad \text{with} \qquad -\frac{1}{y} \leq \vartheta \leq \frac{2}{y}. \tag{A.1}$$

If $c = 1$ the left side is $1/y$ and $\vartheta = 0$. If $c < 1$ we take $x = y$ and $s = 1 - c$. Then the left side lies between $(y+1)^{-c}$ and y^{-c}, and $(1+1/y)^{-c} \geq 1 - c/y \geq 1 - 1/y$. If $c > 1$ we take $x = y$ and $s = 2 - c$ and use $\Gamma(y+1-c) = \Gamma(y+2-c)/(y+1-c)$. Then the left side lies between $(y+1)^{1-c}/(y+1-c)$ and $y^{1-c}/(y+1-c)$. Here $y/(y+1-c) = 1 + (c-1)/(y+1-c) \leq 1 + 1/(y-1) \leq 1 + 2/y$, and $(1+1/y)^{1-c}\,y/(y+1-c) \geq (1+1/y)^{-1} \geq 1 - 1/y$.

A.2 Averaging at the center and a bracket

Lemma A.1 (averaging at the center)**.** *Let $N \geq 2$ be even and $T = T_h^+$. Then*

$$C_N = 2\sum_{r=h}^{N-1} \mathbb{P}(T = r)\,\lambda_{r+1,N} \qquad \textit{and} \qquad \sum_{r=h}^{N-1} \mathbb{P}(T = r) = \frac{1}{2}. \tag{A.2}$$

If $0 < c \leq 1$, then $c/N \leq C_N \leq \lambda_{h+1,N}$.

Proof. On the event $\{A_N = h,\, X_N = -1\}$ the last step equal to $+1$ is the h-th one, at a time $T = r \leq N - 1$. It is followed by a switch at $r+1$ and no switch in $(r+1, N]$. Conversely, these conditions give $A_N = h$ and $X_N = -1$. The event $\{T = r\}$ depends only on $X_1, \chi_2, \ldots, \chi_r$. So $\mathbb{P}(A_N = h,\, X_N = -1) = \sum_{r=h}^{N-1} \mathbb{P}(T = r)\,\lambda_{r+1,N}$, and flip symmetry gives the same value for $X_N = +1$. Next, $T \leq N - 1$ exactly when $A_{N-1} \geq h$. This has

probability $\frac{1}{2}$, because $N-1$ is odd and A_{N-1} is symmetric about $(N-1)/2$. Finally $\lambda_{t+1,N}/\lambda_{t,N} = t/(t+1-c)$, which is at most 1 for $c \le 1$. So $\lambda_{N,N} = c/N \le \lambda_{t,N} \le \lambda_{h+1,N}$ for $h+1 \le t \le N$, and by (A.2) C_N is a weighted average of these values. □

For an integer $M \ge 2$ put $\eta_M(c) = M\rho(1,M)/(2c)$. Since $\rho(1,M)$ decreases in c and equals $1/M$ at $c=1$, the function η_M decreases strictly on $(0,1]$, from ∞ at 0 to $\eta_M(1) = \frac{1}{2}$. Let $c^{(M)} \in (0,1)$ be the root of $\eta_M = 1$.

Proposition A.2 (bracket)**.** *The numbers $c^{(M)}$ increase with M, and $1-c^{(M)} = \frac{\log 2}{\log M+1+\gamma} + O\big((\log M)^{-3}\big)$. For every even $N \ge 2$ we have $c^{(h+1)} \le c_N \le c^{(N)}$. So $(1-c_N)\log N \to \log 2$, and $c_N \ge c^{(3)} = (9-\sqrt{57})/2 > 0.725$ for $N \ge 4$.*

Proof. The equation $\eta_M(c) = 1$ says that $\frac{1}{2}\rho(1,M) = c/M$, namely that the end for M steps equals c/M. We compare E_N with the 2 bounds for C_N in Lemma A.1, and each comparison is an inequality for some η_M.

Monotonicity. For $c<1$ we have $\eta_{M+1}/\eta_M = (M+1-c)/M > 1$. So $\eta_{M+1}(c^{(M)}) > 1$, and since η_{M+1} decreases, $c^{(M+1)} > c^{(M)}$.

The bracket. By Proposition 2.6(b), $C_N < E_N$ for $c < c_N$ and $C_N > E_N$ for $c > c_N$, and $c_N < 1$. So it is enough to compare C_N with E_N for $0 < c < 1$. Let $c^{(N)} < c < 1$. Then $\eta_N(c) < 1$, that is $E_N = \frac{1}{2}\rho(1,N) < c/N$, and $c/N \le C_N$ by Lemma A.1. So $C_N > E_N$, which means $c > c_N$. Hence $c_N \le c^{(N)}$. Now let $c < c^{(h+1)}$. Then $\eta_{h+1}(c) > 1$, that is $\frac{1}{2}\rho(1,h+1) > c/(h+1)$. With $\rho(1,N) = \rho(1,h+1)\rho(h+1,N)$ this gives $E_N > \frac{c}{h+1}\rho(h+1,N) = \lambda_{h+1,N} \ge C_N$. So $c < c_N$. Hence $c_N \ge c^{(h+1)}$.

The expansion. The number $c^{(3)}$ solves $3\rho(1,3) = (2-c)(3-c)/2 = 2c$, that is $c^2 - 9c + 6 = 0$. For $N \ge 4$ we have $h+1 \ge 3$, so $c_N \ge c^{(h+1)} \ge c^{(3)}$. By (A.1) with $y = M$ we have $M\rho(1,M) = M^{1-c}(1+O(1/M))/\Gamma(2-c)$. So the equation $\eta_M(c) = 1$ reads $(1-c)\log M = \Theta_0(c) + O(1/M)$ with $\Theta_0(c) = \log\big(2c\,\Gamma(2-c)\big)$, uniformly for $0 < c \le 1$. For $M \ge 3$ we have $c^{(M)} \in [c^{(3)}, 1)$, where Θ_0 is bounded. So $\varepsilon = 1 - c^{(M)}$ is $O(1/\log M)$, and we may expand at $c = 1$. We have $\Theta_0(1) = \log 2$ and $\Theta_0'(1) = 1 - \psi(1) = 1 + \gamma$, where $\psi = \Gamma'/\Gamma$ and $\psi(1) = -\gamma$ [12, 5.4.12]. Taylor's theorem gives $\varepsilon(\log M + 1 + \gamma) = \log 2 + O(\varepsilon^2) + O(1/M)$, which is the expansion. The limit of $(1-c_N)\log N$ follows since $\log(h+1) \sim \log N$. □

A.3 A local limit at the center

Proposition A.3 (local limit at the center)**.** *For each compact $J \subset (0,2)$ there is C_J such that $\big|NC_N - d_c\big| \le C_J\sqrt{\log N/N}$ for $c \in J$ and even $N \ge 4$.*

Proof. We start from the averaging identity of Lemma A.1 and replace the law of $T = T_h^+$ by its continuum version. Since $T \le r$ exactly when $A_r \ge h$, we need the law of A_r. The chain of Proposition 2.1 with $d = 2$ and $\theta = c$ switches with nearly the same probabilities as the walk, and its counts are binomial with a random parameter B. So A_r is close in law to rB, and the time T is close in law to h/B. The weights $\lambda_{r+1,N}$ are explicit, so the average becomes an explicit integral against the Beta(c,c) density.

The proof has 5 steps. Step 1 shows that the counts of the chain are binomial with the random parameter B. Step 2 couples the walk with the chain and bounds the difference of the distribution functions of T and h/B. Step 3 uses summation by parts to replace the law of T by that of h/B in the average. Step 4 writes the new sum as an expectation over B, and Step 5 evaluates it.

Let $F(r) = \mathbb{P}(T \le r) = \mathbb{P}(A_r \ge h)$, $\Phi(r) = \mathbb{P}(rB \ge h)$ and $\Lambda_r = N\lambda_{r+1,N}$. By Lemma A.1,

$$NC_N = 2\sum_{r=h}^{N-1} \big(F(r) - F(r-1)\big)\Lambda_r.$$

In this proof C depends only on J and may change from line to line. Since $NC_N \le N$ and d_c is bounded on J, it is enough to prove the claim for large N. We put $K = \lceil \sqrt{N} \rceil$, $u = \sqrt{N \log N}$ and $w = K + u$, and we take N so large that $w \le h/4$. Note that $K \le h$.

Step 1 (a chain with binomial counts). Let X' be the chain of Proposition 2.1 with $d = 2$ and $\theta = c$, written with the states ± 1, and let A'_r be its number of steps $+1$ among the first r. Its first step is a fair sign, and at step $k \ge 2$ it switches with probability $q_k = c/(k-1+2c)$, independently of the past. By Proposition 2.1 and $x^{(m)} = \Gamma(x+m)/\Gamma(x)$,

$$\mathbb{P}(A'_r = s) = \binom{r}{s} \frac{c^{(s)} c^{(r-s)}}{(2c)^{(r)}} = \binom{r}{s} \frac{\mathrm{B}(c+s, c+r-s)}{\mathrm{B}(c,c)} = \mathbb{E}\Big[\binom{r}{s} B^s (1-B)^{r-s} \Big].$$

So A'_r has the law of a binomial variable with r trials and the random success probability B. By Hoeffding's inequality [10, Theorem 1], a binomial variable with r trials and success probability x is at least $rx + u$ with probability at most $e^{-2u^2/r}$, and the same holds for being at most $rx - u$. Hence for every real y

$$\mathbb{P}(rB \ge y + u) - e^{-2u^2/r} \le \mathbb{P}(A'_r \ge y) \le \mathbb{P}(rB \ge y - u) + e^{-2u^2/r}.$$

Step 2 (coupling the walk with X'). The switch probabilities differ by $c/k - q_k = c(2c-1)/(k(k-1+2c))$, which is at most $6/(k(k-1))$ in absolute value since $c < 2$. Let $K \le r \le N$. For $K < k \le r$ we couple the switch indicator χ_k of the walk with the switch indicator χ'_k of X' so that $\mathbb{P}(\chi_k \ne \chi'_k) = |c/k - q_k|$, using independent extra randomness. We put $X_K = X'_K$ and $X_k = X_{k-1}(-1)^{\chi_k}$ for $K < k \le r$, and we draw $X_1, \dots, X_{K-1}$ from the law of the walk given X_K, again with independent randomness. In both processes the sign at step K is fair, and the later switches are independent of the first K steps. So $(X_1, \dots, X_r)$ has the law of the walk. Outside an event of probability at most $\sum_{k>K} 6/(k(k-1)) = 6/K$ the two processes agree at the steps $K, \dots, r$, and then $|A_r - A'_r| < K$. With Step 1, $w = K + u$ and $e^{-2u^2/r} \le e^{-2\log N}$, this gives

$$|F(r) - \Phi(r)| \le \mathbb{P}(h - w \le rB < h + w) + \frac{1}{N^2} + \frac{6}{K} \qquad (K \le r \le N).$$

Step 3 (replacing F by Φ). Put $\Delta(r) = F(r) - \Phi(r)$. Then $\Delta(h-1) = 0$, because $A_{h-1} < h$ and $B < 1$. Summation by parts gives

$$\sum_{r=h}^{N-1} \big(\Delta(r) - \Delta(r-1)\big) \Lambda_r = \Delta(N-1)\Lambda_{N-1} - \sum_{r=h}^{N-2} \Delta(r) \big(\Lambda_{r+1} - \Lambda_r\big).$$

The ratio $\lambda_{t+1,N}/\lambda_{t,N} = t/(t+1-c)$ gives $\Lambda_{r+1} - \Lambda_r = \Lambda_r (c-1)/(r+2-c)$. Every factor of $\rho(r+1, N)$ lies in $(0,1)$, so $\Lambda_r = \frac{Nc}{r+1}\rho(r+1,N) \le 2c$ for $r \ge h$. Hence $\Lambda_{N-1} = c$ and $|\Lambda_{r+1} - \Lambda_r| \le 2c/h \le 8/N$, and the right side is at most $c|\Delta(N-1)| + \frac{8}{N}\sum_{r=h}^{N-2} |\Delta(r)|$ in absolute value. For each value b of B the number rb lies in $[h-w, h+w)$ for at most $2w/b + 1$ integers r, and for no $r \le N$ unless $b \ge (h-w)/N \ge \frac{3}{8}$. So the probabilities in Step 2 add up to at most $6w + 1$ over r, and $\sum_{r=h}^{N-2} |\Delta(r)| \le 6w + 2 + 3\sqrt{N}$. For $r = N-1$ the event in Step 2 puts B in an interval of length $2w/(N-1)$ inside $[\frac{3}{8}, \frac{3}{4}]$, where the density of B is at most C for $c \in J$. So $|\Delta(N-1)| \le Cw/N + N^{-2} + 6/K$. Together, replacing F by Φ changes NC_N by at most $C(w/N + 1/K) \le C\sqrt{\log N / N}$.

Step 4 (the new sum is an expectation). After the replacement the sum is

$$2 \sum_{r=h}^{N-1} \big(\Phi(r) - \Phi(r-1)\big) \Lambda_r.$$

The term with $r = h$ vanishes, since $\Phi(h) = \Phi(h-1) = 0$. For $h < r \le N-1$ we have $\Phi(r) - \Phi(r-1) = \mathbb{P}(h/r \le B < h/(r-1))$, and on this event $\lceil h/B \rceil = r$. These

events are disjoint, and their union is $\{h/(N-1) \le B < 1\}$. So the new sum is $2\,\mathbb{E}\big[\Lambda_{\lceil h/B\rceil};\, B \ge h/(N-1)\big]$.

Step 5 (evaluating the expectation). By (A.1) with $y = N$ and $y = r+1 \ge 3$, $\Lambda_r = c\big((r+1)/N\big)^{c-1}(1+O(1/N))$ for $h \le r \le N-1$. On the event above with $r = \lceil h/B\rceil$, the number $(r+1)/N$ lies in $[\frac12, 2]$ and within $2/N$ of $h/(BN) = 1/(2B)$, which also lies in $[\frac12, 2]$. The function x^{c-1} is 4-Lipschitz on $[\frac12, 2]$. So $\Lambda_{\lceil h/B\rceil} = c(2B)^{1-c} + O(1/N)$. Also $\mathbb{P}\big(\frac12 \le B < h/(N-1)\big) = O_J(1/N)$, since the density of B is bounded near $\frac12$ uniformly on J. So we may replace the event $\{B \ge h/(N-1)\}$ by $\{B \ge \frac12\}$. In the remaining integral the factor b^{1-c} cancels the factor b^{c-1} of the density. Hence, by Step 3, NC_N is within $C\sqrt{\log N/N}$ of

$$2c\,\mathbb{E}\big[(2B)^{1-c};\, B \ge \tfrac12\big] + O_J(1/N) = \frac{2c\,2^{1-c}}{\mathrm{B}(c,c)} \int_{1/2}^{1} (1-b)^{c-1}\, db + O_J(1/N) = d_c + O_J(1/N). \quad \square$$

A.4 The expansion of the crossing

Proof of Proposition 2.7. We write $\log C_N - \log E_N$ as an explicit function of c with a small error, and we solve for its zero near $c = 1$. Let $J = [c^{(3)}, 1]$. Then d_c is bounded below on J by a positive constant. By (A.1) with $y = N$, $\log E_N = \log\big(\frac12 \rho(1,N)\big) = -\log 2 - \log\Gamma(2-c) - c\log N + O(1/N)$. By Proposition A.3, $\log C_N = \log d_c - \log N + O(\sqrt{\log N/N})$ for large N. Both hold uniformly for $c \in J$, so

$$\log C_N - \log E_N = (c-1)\log N + \Theta(c) + O(\sqrt{\log N/N}) \qquad (c \in J). \tag{A.3}$$

Here $\Theta(1) = \log 2$ because $d_1 = 1$. Next, $\Theta'(c) = -\psi(2-c) - \log 4 - 2\psi(c) + 2\psi(2c)$, and with $\psi(1) = -\gamma$ and $\psi(2) = 1 - \gamma$ [12, 5.4.12, 5.4.14] this gives $\Theta'(1) = \gamma + 2 - 2\log 2$. Put $L = \log N + \Theta'(1)$. By Proposition A.2, $c_N \in J$ for $N \ge 4$ and $\varepsilon = 1 - c_N$ is $O(1/\log N)$. The left side of (A.3) vanishes at $c = c_N$, because $C_N = E_N$ there. As Θ is smooth on J, Taylor's theorem gives $\Theta(c_N) = \log 2 - \varepsilon\,\Theta'(1) + O(\varepsilon^2)$. So (A.3) at $c = c_N$ reads $0 = -\varepsilon\log N + \log 2 - \varepsilon\,\Theta'(1) + O(\varepsilon^2) + O(\sqrt{\log N/N})$, that is $\varepsilon L = \log 2 + O(\varepsilon^2) + O(\sqrt{\log N/N}) = \log 2 + O(L^{-2})$. So $\varepsilon = \log 2/L + O(L^{-3})$, which is the expansion. $\square$

With one more term in Taylor's theorem and $\varepsilon^2 = (\log 2)^2/L^2 + O(L^{-4})$, the same argument adds the term $-\Theta''(1)(\log 2)^2/(2L^3)$ to the expansion of c_N, with error $O(L^{-4})$. Here $\Theta''(1) = \pi^2/2 - 4$, since $\Theta''(c) = \psi'(2-c) - 2\psi'(c) + 4\psi'(2c)$, $\psi'(1) = \zeta(2)$ and $\psi'(2) = \zeta(2) - 1$ [12, 5.4.12, 5.15.5]. At $N = 200$ the bracket of Proposition A.2 is $[0.88749, 0.89878]$, Proposition 2.7 gives 0.89319, and with the L^{-3} term we get 0.89236. The high-precision value is $c_{200} = 0.89253$. The constants C_J and the constants in the O-terms of this appendix are not computed. Numerically $|NC_N - d_c|$ decays like $1/N$. At $c = 0.75$ it drops by a factor of about 4 each time N is multiplied by 4, from $N = 400$ to $N = 6400$. So the rate in Proposition A.3 is probably not sharp.

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